

10.1 DC: Practical Circuits & Kirchhoff's Laws

Question Paper

Course	CIE A Level Physics
Section	10. D.C. Circuits
Topic	10.1 DC: Practical Circuits & Kirchhoff's Laws
Difficulty	Hard

Time allowed: 20

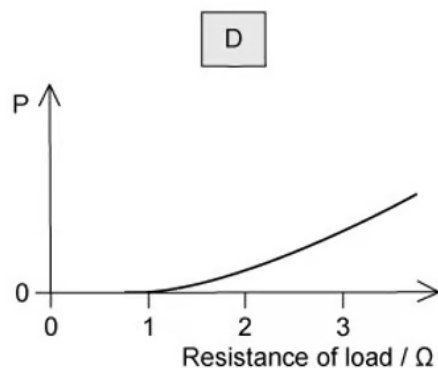
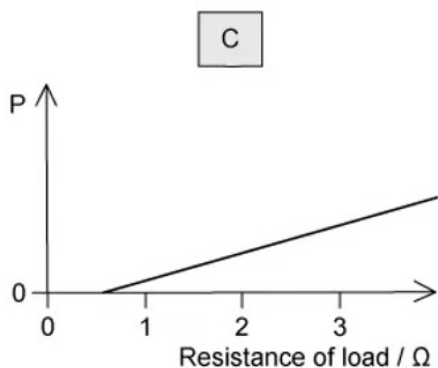
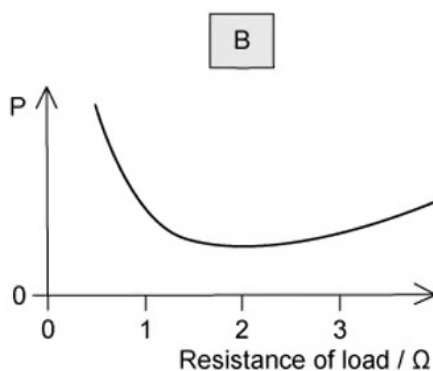
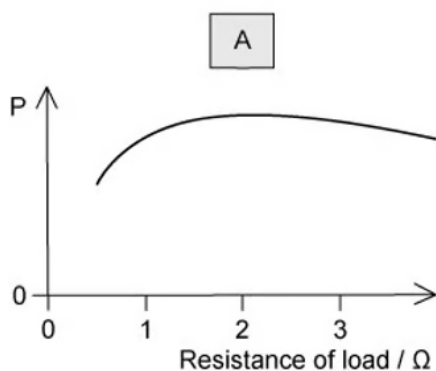
Score: /10

Percentage: /100

Question 1

A power supply of electromotive force (e.m.f.) 12 V and internal resistance $2\ \Omega$ is connected in series with a load resistor. The value of the load resistor is varied from $0.5\ \Omega$ to $4\ \Omega$.

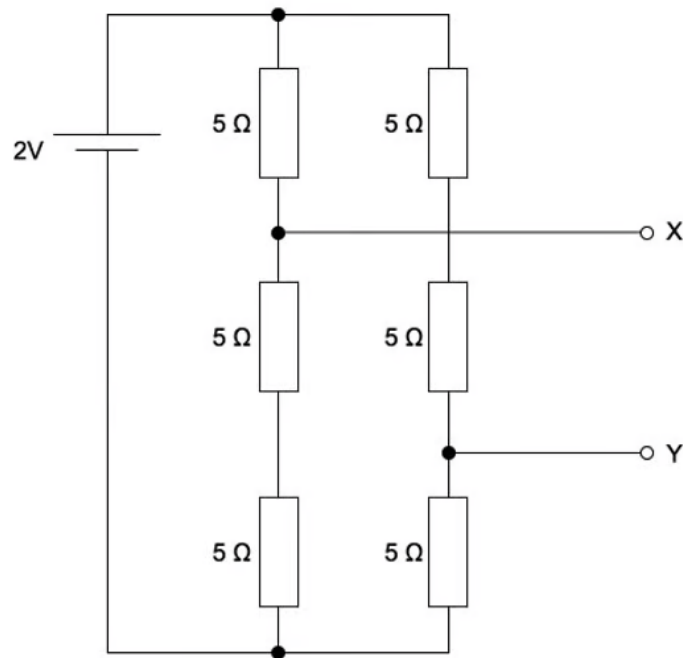
Which graph shows how the power P dissipated in the load resistor varies with the resistance of the load resistor?



[1 mark]

Question 2

Six resistors, each of resistance $5\ \Omega$, are connected to a 2 V cell of negligible internal resistance.



What is the potential difference between terminals X and Y?

- A. $\frac{2}{3}\text{ V}$
- B. $\frac{8}{9}\text{ V}$
- C. $\frac{4}{3}\text{ V}$
- D. 2 V

[1 mark]

Question 3

A relay is required to operate 800 m from its power supply. The power supply has negligible internal resistance. The relay requires 16.0 V and a current of 0.60 A to operate.

A cable connects the relay to the power supply, and two of the wires in the cable are used to supply power to the relay.

The resistance of each of these wires is $0.0050 \, \Omega$ per meter.

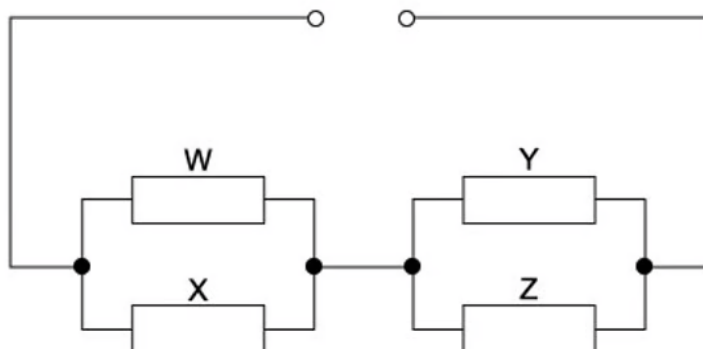
What is the minimum output e.m.f. of the power supply?

- A. 16.6 V
- B. 18.4 V
- C. 20.8 V
- D. 29.3 V

[1 mark]

Question 4

Four resistors of equal value are connected as shown.



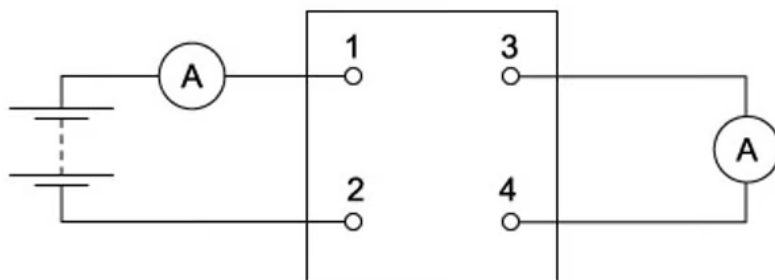
How will the powers to the resistors change when resistor W is removed?

- A. the powers to X, Y and Z will all increase
- B. the power to X will decrease and the powers to Y and Z will increase
- C. the power to X will increase and the powers to Y and Z will decrease
- D. the power to X will increase and the powers to Y and Z will remain unaltered

[1 mark]

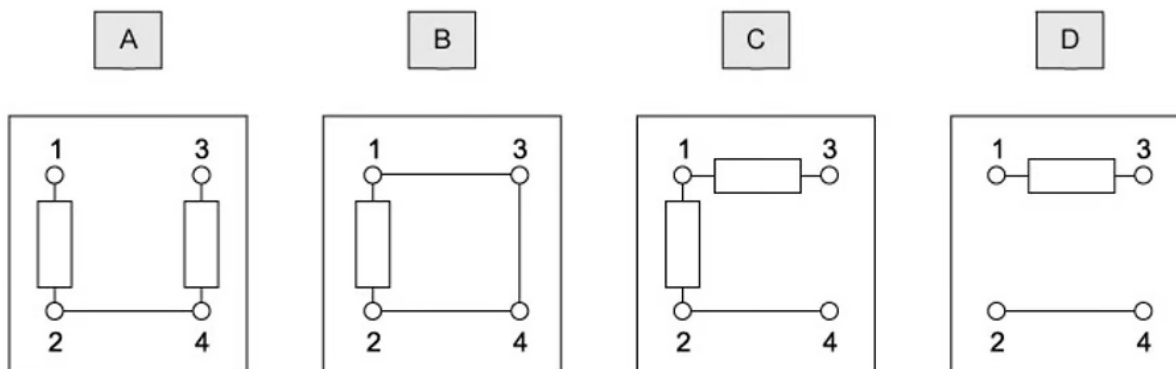
Question 5

The diagram shows a four-terminal box connected to a battery and two ammeters.



The currents in the two meters are identical.

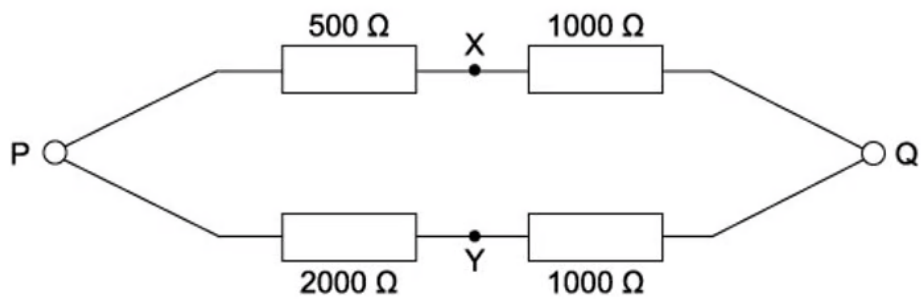
Which circuit, within the box, will give this result?



[1 mark]

Question 6

A p. d. of 12 V is connected between P and Q.



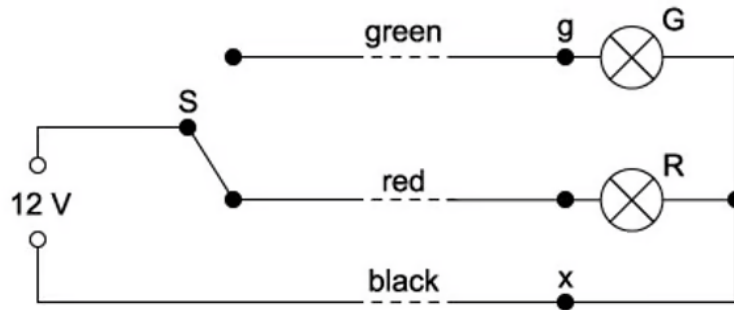
What is the p. d. between X and Y?

- A. 0 V
- B. 4 V
- C. 6 V
- D. 8 V

[1 mark]

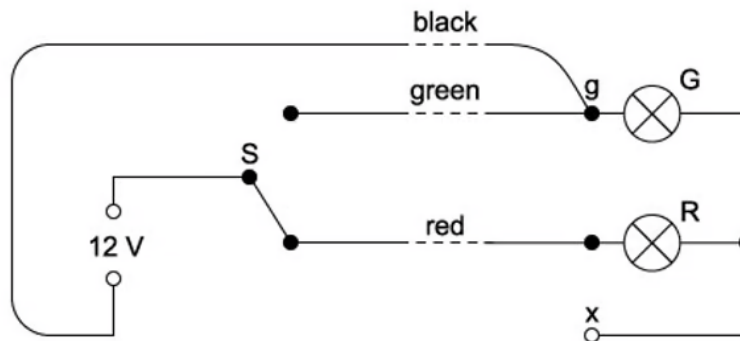
Question 7

The diagram shows the circuit for a signal to display a green or a red light. It is controlled by the switch S.



The signal is some way from S to which it is connected by a cable with green, red and black wires. At the signal, the green and red wires are connected to the corresponding lamp and the black wire is connected to a terminal x to provide a common return. The arrangement is shown correctly connected and with the switch set to illuminate the red lamp.

During maintenance, the wires at the signal are disconnected and, when reconnected, the black wire is connected in error to the green lamp (terminal g) instead of terminal x. The red wire is connected correctly to its lamp and connections at S remain as in the diagram.



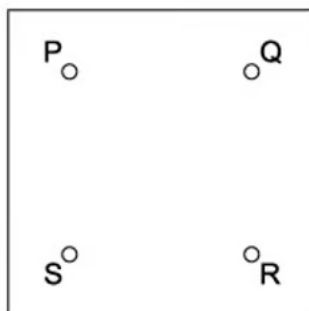
When the system is tested with the switch connection to the red wire, what does the signal show?

- A. the green lamp illuminated normally
- B. the red lamp illuminated normally
- C. the red and green lamps both illuminated normally
- D. the red and green lamps both illuminated dimly

[1 mark]

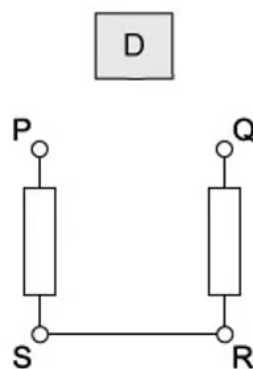
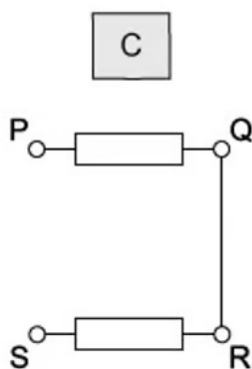
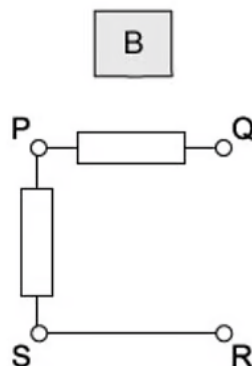
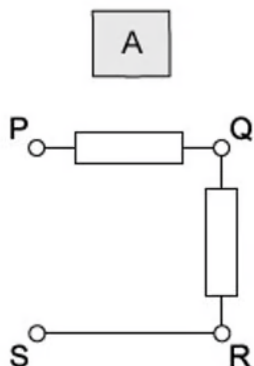
Question 8

A box with four terminals P, Q, R and S contains two identical resistors.



When a battery of electromotive force (e.m.f.) E and negligible internal resistance is connected across PS, a high-resistance voltmeter connected across QR reads $\frac{E}{2}$

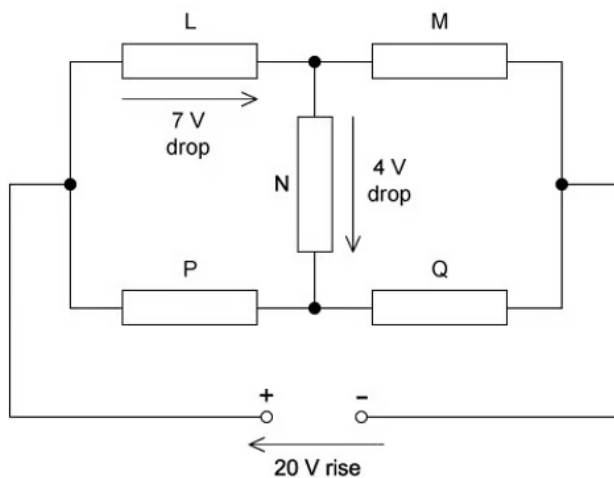
Which diagram shows the correct arrangement of the two resistors inside the box?



[1 mark]

Question 9

A 20 V d. c. supply is connected to a circuit consisting of five resistors L, M, N, P and Q.



There is a potential drop of 7 V across L and a further 4 V potential drop across N.

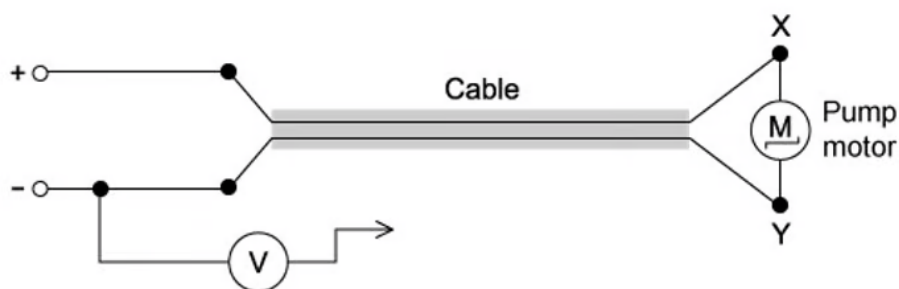
What are the potential drops across M, P and Q?

	Potential drop across M / V	Potential drop across P / V	Potential drop across Q / V
A	9	7	13
B	13	7	13
C	13	11	9
D	17	3	17

[1 mark]

Question 10

The diagram shows the electric motor for a garden pump connected to a 24 V power supply by an insulated two-core cable.



The motor does not work so, to find the fault, the negative terminal of a voltmeter is connected to the negative terminal of the power supply, and its other end is connected in turn to terminals X and Y at the motor.

Which row represents two readings and a correct conclusion?

	Voltmeter reading when connected to X / V	Voltmeter reading when connected to Y / V	Conclusion
A	24	0	break in positive wire of cable
B	24	12	break in negative wire of cable
C	24	24	break in connection within the motor
D	24	24	break in negative wire of cable

[1 mark]



Model Answers: Hard

1

The correct answer is **A** because:

- The internal resistance of the supply is $r = 2\ \Omega$
- The resistance of the load resistor, R varies from $0.5\ \Omega$ to $4\ \Omega$
- Power P dissipated in the load resistor is equal to $I^2 R$
- The power dissipated is maximum when the load resistance is equal to the internal resistance
- This makes sense since when the resistance is the highest, the maximum energy is needed to drive charge through the cell, and hence the power dissipated is maximum (as energy = $\frac{\text{power}}{\text{time}}$)
- On the graph, when the resistance of the load is equal to the internal resistance of $2\ \Omega$ then the curve is at a maxima, as illustrated in option **A**

B is incorrect because this graph shows that the power reaches a minimum value when the load resistance equals the internal resistance

C & D are incorrect because at a glance of the equation between power and resistance $P = I^2 R$, you might initially think the relationship is directly proportional. However, we have to consider the internal and external resistors as two resistors in series. Initially, the value of R is zero, so the power dissipated across it is also zero. Similarly, if the value of R is infinite, the power dissipated is also zero as the current will then also be zero. So, the power dissipated must increase and then decrease as R increases from zero. As the value of R increases, the current I decreases and hence the p.d. across the external resistor increases. The p.d. across the internal resistor r decreases and the power dissipated in r (given by $I^2 r$) decreases. Therefore, the maximum power dissipated across R occurs when $R = r$

2

The correct answer is **A** because:

- To obtain the potential difference between terminals X and Y, we first need to know the values of the potential at X and Y
 - The positive terminal of the cell can be taken to be at a potential of 2 V and the negative terminal to be at a potential of 0 V
- It can be seen that the circuit consists of 2 loops, with a junction at the top and one at the bottom
 - Since there are no components connected between the upper junction and the positive terminal of the cell, the upper junction must be at a potential of 2 V, and the lower junction must be at a potential of 0 V
- Therefore, in any loop, going from up to down, the potential decreases across each component until it is zero at the lower junction
- From Kirchhoff's Law, the sum of p.d. in any loop is equal to the e.m.f in the circuit
- Consider the first loop (where terminal X is connected):
- From the potential divider equation:
 - p.d. across the upper resistor = $\left(\frac{R_1}{R_1 + R_2 + R_3}\right) \times E = \left(\frac{5}{5+5+5}\right) \times 2 = \frac{2}{3} \text{ V}$
 - Note that this is the potential difference across the upper resistor. It is also the p.d. across each resistor since they all have the same resistance
- Since the upper junction is at a potential of 2 V, the p.d. at terminal X is equal to $= 2 - \frac{2}{3} = \frac{4}{3} \text{ V}$
- Consider the second loop (where terminal Y is connected):
- From the upper junction to terminal Y, there are 2 resistors and the p.d. across each is $\frac{2}{3} \text{ V}$
 - So, the p.d. at terminal Y = $2 - 2\left(\frac{2}{3}\right) = \frac{2}{3} \text{ V}$
- Therefore, the p.d. between X and Y = $\frac{4}{3} - \frac{2}{3} = \frac{2}{3} \text{ V}$

3

The correct answer is **C** because:

- In the cable, two of the wires are used to supply power to the relay
- So, the circuit is connected as follows:
 - One terminal of power supply — 1st wire — relay — 2nd wire — other terminal

of power supply, therefore, it is a series connection

- The resistance of each of these wires is 0.0050Ω per meter and each wire measures 800 m long
 - So, the total resistance of one wire $= 800 \times 0.0050 = 4 \Omega$
 - Therefore, the total resistance due to both wires $= 4 + 4 = 8 \Omega$
- There will be a p.d. across the wires and since this is a series connection, the same current flows through the whole circuit
 - So, the p.d. across the two wires $= IR = 0.60 \times 8 = 4.8 \text{ V}$
- The sum of p.d. across the whole circuit should be equal to the minimum e.m.f.
 - Therefore, the minimum output e.m.f. $= 16.0 + 4.8 = 20.8 \text{ V}$

4

The correct answer is **C** because:

- Power dissipated in a resistor $= I^2 R$
- Let's say the resistance of each resistor is R
- Two resistors in series causes the overall resistance to increase: $R + R = 2R$
- Two resistors in parallel causes the overall resistance to decrease :

$$\left[\frac{1}{R} + \frac{1}{R} \right]^{-1} = \frac{R}{2}$$
- Since the 4 resistors have the same resistance in the connection shown, the p.d. across each parallel connection will be the same
 - Overall the resistance in the circuit is $= \frac{R}{2} + \frac{R}{2} = R$
- Let's say the e.m.f. in the circuit is E , so, the current in the circuit is $I = \frac{E}{R}$
- The total current splits at the parallel combination
 - Power dissipated in X $= \left(\frac{1}{2} \right)^2 R = \frac{I^2 R}{4} = 0.25 I^2 R$
 - This is the power dissipated in all the resistors
- When W is removed, the overall resistance of the parallel combination of W and X changes from $0.5R$ to R as only X is left
- Total resistance in new circuit $= R + 0.5R = 1.5R$
 - Current in circuit $= \frac{E}{1.5R} = \frac{2}{3} I$
- The overall current in the circuit will also change — it decreases since the overall resistance in the complete circuit has increased. This current flows through X but is split into half at the combination of Y and Z, so, the current through Y and Z is

half that through X, so:

- Power dissipated in X = $\left(\frac{2}{3}I\right)^2 R = \frac{4}{9}I^2 R = 0.44 I^2 R$

- Half the current flows through Y and Z, and the same power is dissipated through both resistors, so:

- Power dissipated in Y = $\left[\frac{1}{2}\left(\frac{2}{3}I\right)\right]^2 R = \frac{1}{9}I^2 R = 0.11 I^2 R$

- Therefore, the power in X increases and the power in Y and Z decreases

5

The correct answer is **D** because:

- The ammeter connected close to the battery gives the total current in the circuit while the ammeter connected between terminals 3 and 4 gives the current passing through terminal 3 to 4
- The current in this one will only be equal to the one close to the battery only if all components are connected in series in the whole circuit, including within the box
- Consider circuit **D**:
 - The 2 ammeters are in series with the resistor between terminals 1 and 3
 - So, the same current flows through components connected in series
 - Therefore, option **D** is correct

A is incorrect because the section from terminal 2 to 4 and finally to 3 is short-circuited (since 1 and 3 are not connected), so, no current flows through the ammeter

B is incorrect because the ammeter connected to 3 and 4 is short-circuited since the wires (which have negligible resistance) are connected between 1 to 3, 3 to 4 and 4 to 2

C is incorrect because a parallel connection is formed between the 2 resistors between 1 to 3 and 1 to 2, so, the current splits up at junction 1. The second ammeter is connected in series with one of the resistors, so it only gives the current flowing in one resistor

6

The correct answer is **B** because:

- The statement "p. d. of 12 V is connected between P and Q" means that the

⚡ is incorrect because a parallel connection is formed between the 2 resistors between 1 to 3 and 1 to 2, so, the current splits up at junction 1. The second ammeter is connected in series with one of the resistors, so it only gives the current flowing in one resistor

6

The correct answer is **B** because:

- The statement "p. d. of 12 V is connected between P and Q" means that the difference in potential between P and Q is 12 V
- So, any potential can be assigned to P and Q as long as the difference is 12 V
 - Let's assign a potential of 0 V at P and 12 V at Q
- So, when we go from P to Q, the potential should increase and point X would be at a potential between 0 and 12 V
- The same is true for point Y; hence, it can be concluded at points P, Q, X and Y are all at different potentials
- The circuit consists of a parallel combination of resistors
 - So, the total p. d. across the upper and lower parts of the circuit are both 12 V
- Now, both the upper and lower part of the circuit each consist of two resistors in series
- For series connection of two resistors R_1 and R_2 , the p. d. across R_1 is given by the potential divider equation
 - $V = \left(\frac{R_1}{R_1 + R_2} \right) \times E$ where E is the e. m. f. of the supply
- The potential at point X = p. d. across 500 Ω in the upper part
 - Potential at point X = $\left(\frac{500}{500 + 1000} \right) \times 12 = 4 \text{ V}$
- Similarly, the potential at point Y = p. d. across 2000 Ω in the upper part
 - Potential at point Y = $\left(\frac{2000}{2000 + 1000} \right) \times 12 = 8 \text{ V}$
- So, the p. d. between X and Y is the difference in potential at point X and point Y
 - Therefore, p. d. between X and Y = $8 - 4 = 4 \text{ V}$

Exam Tip: Note that 0 V could be assigned at Q and 12 V at P, but the process discussed would have to be reversed

7

The correct answer is **D** because:

- The lamps are both lit dimly as they are connected in series
- From Kirchhoff's First Law: the same power supply is now providing the same current to both lights at the same time
- Ohm's Law states $V = IR$
 - Since the total resistance in the circuit is higher and the current flowing in the circuit is less, as $R = \frac{V}{I}$
- Each of the lights now receive a voltage lower than 12 V as the total voltage is now shared between the two lamps
- Previously, all the 12 V was being provided to a single lamp which would cause it to be brighter

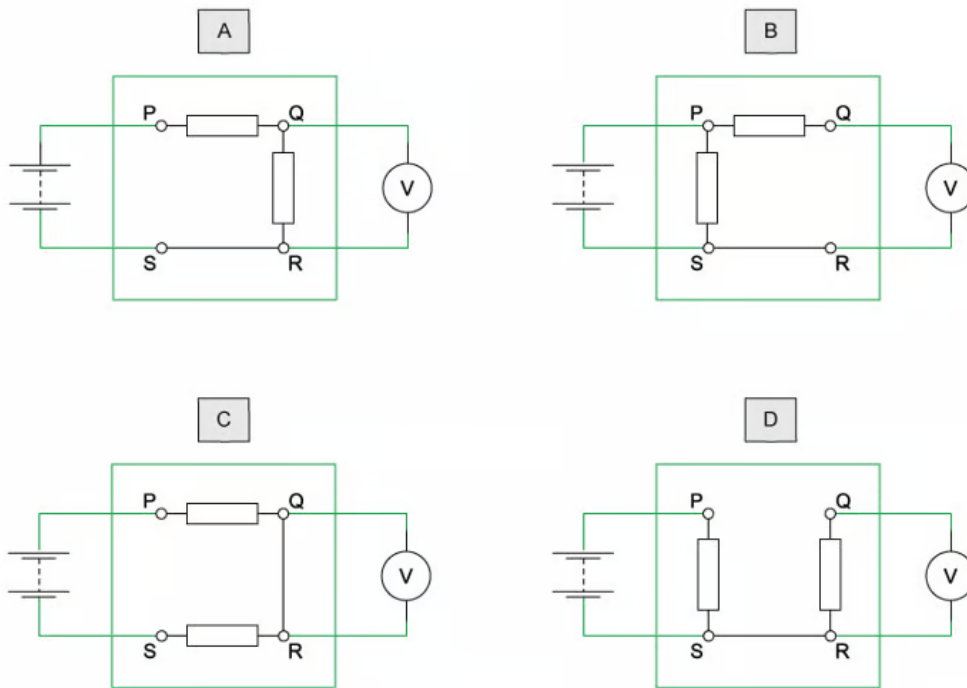
A & B are incorrect because both lamps are connected in series; hence they will both be lit

C is incorrect because both lamps are connected in series, but now the voltage is shared between them; hence they cannot have the same brightness as the single lit lamp previously

8

The correct answer is **A** because:

- The best approach to this question is to draw the battery and voltmeter in each of the four circuits, as shown below
- The reading on the voltmeter across QR is $\frac{E}{2}$
- Ohm's Law states $V = IR$
 - This means that half of the resistance, or one resistor, is across QR
- Consider circuit **A**:
 - The voltmeter is connected across one of the two resistors, and the circuit is complete
 - The reading on the voltmeter would therefore be $\frac{E}{2}$
 - So, option A is correct



B & C are incorrect because to get a reading of $\frac{E}{2}$ there would need to be just one resistor across QR

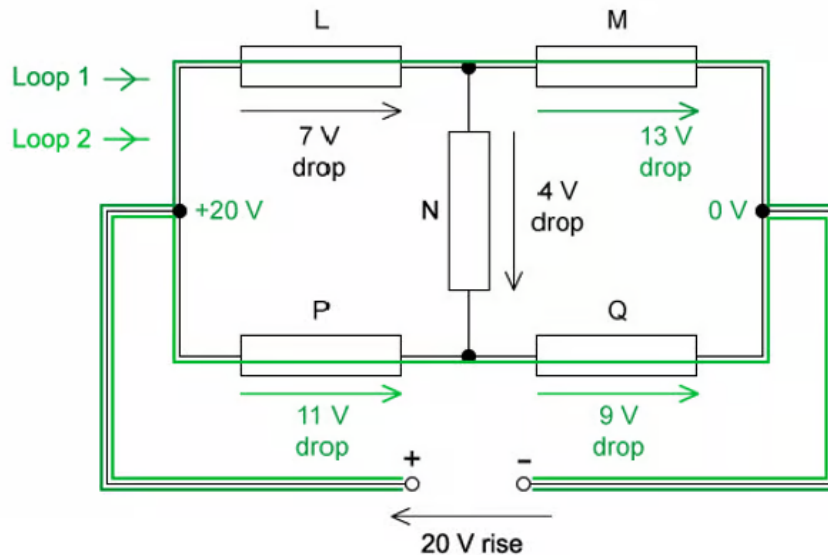
D is incorrect because this is not a complete circuit, so the reading would be zero

9

The correct answer is **C** because:

- This question can be tackled by considering the different loops present in the circuit and applying Kirchhoff's Laws to them
- Consider loop 1: (+) terminal $\rightarrow$ resistor L $\rightarrow$ resistor M $\rightarrow$ (-) terminal
- Kirchhoff's Second Law states: the sum of p. d. across any loop should be equal to the e. m. f.
- So, a 7 V drop across resistor L means a 13 V drop across M as $20 - 7 = 13$ V
 - The junction between L and M is at a potential of 13 V, since the right junction to which M is connected is at 0 V as it is connected to the (-) terminal of the supply, so the sum of p. d. across L and M = 20 V
- Consider loop 2: (+) terminal $\rightarrow$ resistor P $\rightarrow$ resistor Q $\rightarrow$ (-) terminal
- There is a potential drop of 4 V downwards across N, so current flows downwards

- This means that the potential through loop 1 is greater than through loop 2 and the difference in potential is 4 V
 - Therefore, loop 2 (between P and Q) is at a potential of $13\text{ V} - 4\text{ V} = 9\text{ V}$
- So, the potential drop across resistor Q is 9 V
- Finally, consider resistor P: its terminal is at a potential of 20 V and its right terminal is at a potential of 9 V
 - Therefore, the potential drop across P = $20 - 9 = 11\text{ V}$



10

The correct answer is **D** because:

- From the diagram, X is connected to the positive terminal of the power supply, and Y is connected to the negative terminal of the power supply
- So, if the potential at X is 24 V, which it is for all options, then there must be no break in the positive wire
- Now, consider a break in the negative wire of the cable
- The circuit is not complete, and so, there is no current passing through the circuit
- So, there is no drop in potential across the pump motor (in other words, the potential at Y is the same as the potential at X)
- Therefore, a break in the negative wire of the cable would give a potential of 24 V at Y, as well as 24 V at X

A is incorrect because if there was a break in the positive wire, the voltmeter reading when connected to X or Y would be zero

B is incorrect because if there was a break in the negative wire, the potential at Y would be the same as the potential at X (24 V)